Comparing Univariate ANOVA and MANOVA for Hypothesis Testing in Cyclic Data Settings*

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1 Introduction

Real world data is rarely perfectly normally distributed. Though many statistical models require assumptions to hold true for inference, there is still valuable information extracted from imperfect data. Especially in nature, patterns often surface when considering biological and seasonal phenomena. For example, data extracted ranging from the angular velocity of zebrafish in circular wells to heart rhythm monitoring can lead to predictable cycles. According to past ecology research paper, "The Multivariate Analysis of Variance as a Powerful Approach for Circular Data" (Landler, Ruxton, and Malkemper (2022)), this circular data can be statistically addressed using Multivariate Analysis of Variance (MANOVA) to broaden the scientific applications.

When observations within a single population follow a trend, independence is violated, leading to inflated type one error rates along with biased p-values. To extend beyond Lander et al.'s research, we will observe how cyclic data can impact the hypothesis testing performance of MANOVA compared to univariate ANOVA. In order to do so, we will simulate 3-dimensional data for two distinct populations using R. For our first experiment, we test if violations to the independence assumption of cyclic data can still detect a true difference in population means. Our second question of interest pertains to whether the same mean vectors will register as so when we add sufficient noise to negate the cyclic dependency. In both scenarios, we compare the results between the MANOVA output with the majority hypothesis decision for the univariate ANOVA tests, corrected using Bonferroni for multiple testing.

Findings indicate that MANOVA is considerably superior to ANOVA at detecting a true difference in means while performance is comparable between methods when there is addition of enough noise to mask data dependency. We corroborate that MANOVA is adequate at

*Project repository available at: https://github.com/christiecngo/math257-project1.

detecting differences in means for circular data despite its independence violation. Future areas of study include differing the length of cycles between populations and incorporating errors in only some dependent responses.

2 Method

MANOVA is a model for comparison of multivariate mean vectors. The observation vector takes the form:

$$X_{lj} = \mu + \tau_l + e_{lj}$$

This notation references Johnson and Wichern (2007). l is the number of observations in each group whereas j represents the number of groups. Each component in X_{lj} satisfies the univariate ANOVA model. μ is the overall mean while τ_l is the treatment effect of the lth treatment subject to the constraint $\sum_{l=1}^{g} n_l \tau_l = 0$ for identifiability. $\mathbf{e}_{lj}$ are errors represented by independent variables of distribution $N_p(0, \Sigma)$.

AS an extension of ANOVA, MANOVA differs from univariate because the method tests multiple variables simultaneously instead of aggregating results at the end. We expect MANOVA to perform more consistently as there is a lower chance of obtaining spurious results than when we observe univariate dependent variables.

2.1 Assumptions

- Both populations are distributed as multivariate normal (univariate normal for ANOVA).
- Observations are randomly sampled from their respective population and independent within and between groups.
- The two groups share a common variance (Σ matrix in MANOVA).

2.2 Hypothesis Testing

We let the baseline population be 1 and the modified population be 2. Our testing is as followed:

ANOVA (for each of the dimensions):

$$H_0 : \mu_1 = \mu_2$$

$$H_a : \mu_1 \neq \mu_2$$

MANOVA:

$$H_0 : \vec{\mu}_1 = \vec{\mu}_2$$

$$H_a : \vec{\mu}_1 \neq \vec{\mu}_2$$

2.3 Multiple Testing Correction

To account for the repeated univariate tests, we apply the Bonferroni correction by dividing our $\alpha = 0.05$ value by the number of tests (3). Because increasing the number of tests allows for more opportunity to obtain a significant result due to chance, we account for this by using a more conservative p-value threshold.

2.4 Technology

Data simulations and hypothesis testing were all conducted using R software (R Core Team (2026)). The following packages were used: mvtnorm, tidyverse, and scatterplot3d. ChatGPT (OpenAI (2026)) was used for formatting tables and figures as well as some conceptual verification.

2.5 Simulations

In both generated datasets, we create two populations with three dimensions and 200 observations. The independent vector x consists of 200 points equally spaced across 32π or 16 rotations (with the addition of some normally distributed noise centered at 0 with a low standard deviation of 0.1). Each observational unit is a data point within the rotation. The dependent vectors $Y1$, $Y2$, and $Y3$ apply the sine of x function, the cosine of x function, and sine of $2x$ function, respectively. Without any modifications, the mean value of each function should be symmetric with respect to the x-axis and thus have an expected value of 0.

2.5.0.1 Cyclic Dependency

To visually assess the strength of the cyclic relationship, we use the Autocorrelation Function (ACF) plot. At each lag, which comprises of data at past times in the series, we are able to determine its correlation with the series. Repeating peaks at predictable time intervals indicate seasonality. The blue dotted bands represent a 95% confidence limit, which tests for correlations within observations. If spikes fall within the bands, then it can be treated as random noise.

2.5.0.2 Simulation 1: True difference in means

We want to maintain the cyclic structure, so we add a small amount of normally distributed noise but with a slightly shifted mean of 0.12. The differences are apparent in the lack of overlap between the populations in Figure 1. Using the Figure 2 plot below, we can confirm that there is still seasonality in our generated dataset.

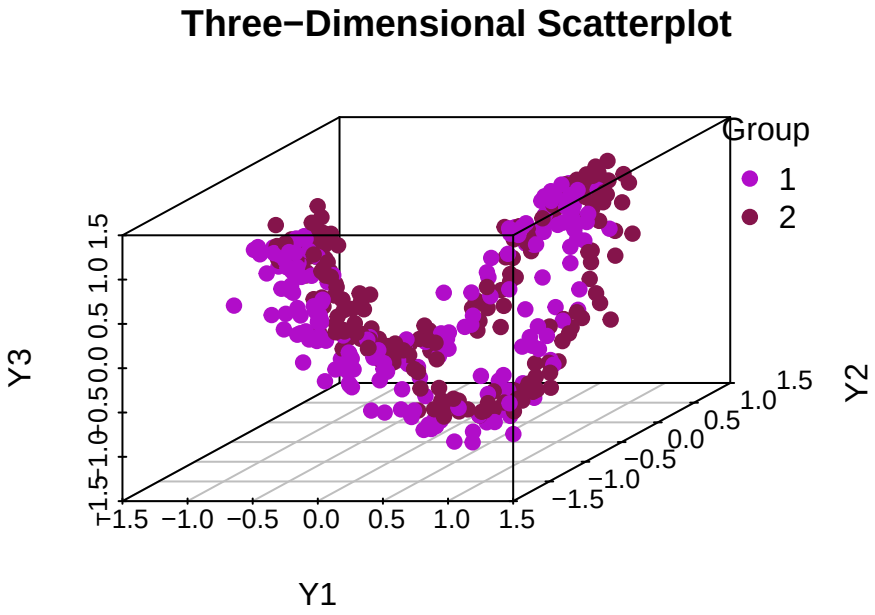


Figure 1: 3D Scatterplot of True Difference In Cyclic Means

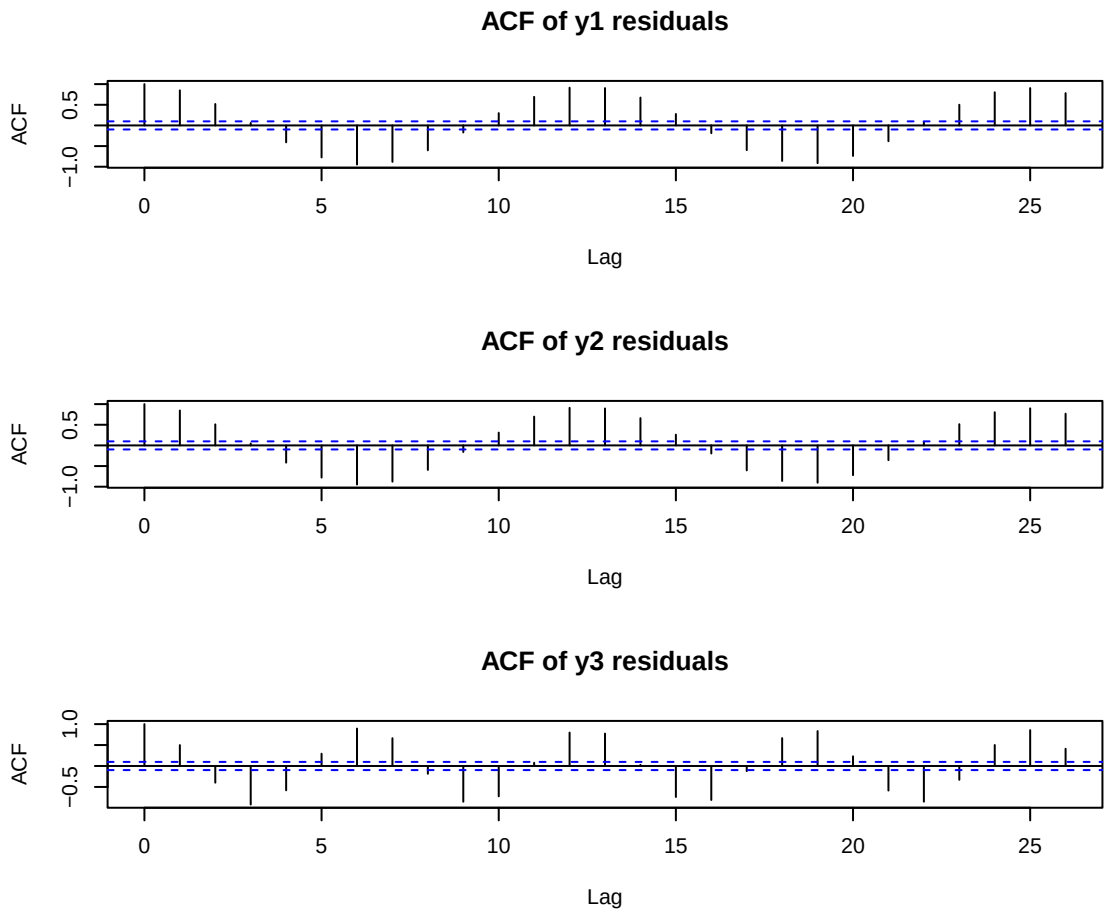


Figure 2: ACF Plot of Each Dimension's Residuals for True Difference In Cyclic Means

2.5.0.3 Simulation 2: Independence from cyclic data

For our second simulation, we want to see if adding sufficient noise to effectively reduce dependence between points in the cycle will lead to hypothesis tests detecting significant differences between populations, despite them being identically distributed. We started with a standard deviation of 0.5 and tested incremental increases until the independence assumption held with standard deviation equal to 1. In the Figure 3 plot, we can see that the noise now makes the populations appear indistinguishable, which is promising given their identical distributions. Again, we examine the ACF plot (Figure 4) to confirm that the majority of spikes are well within the blue dotted bands.

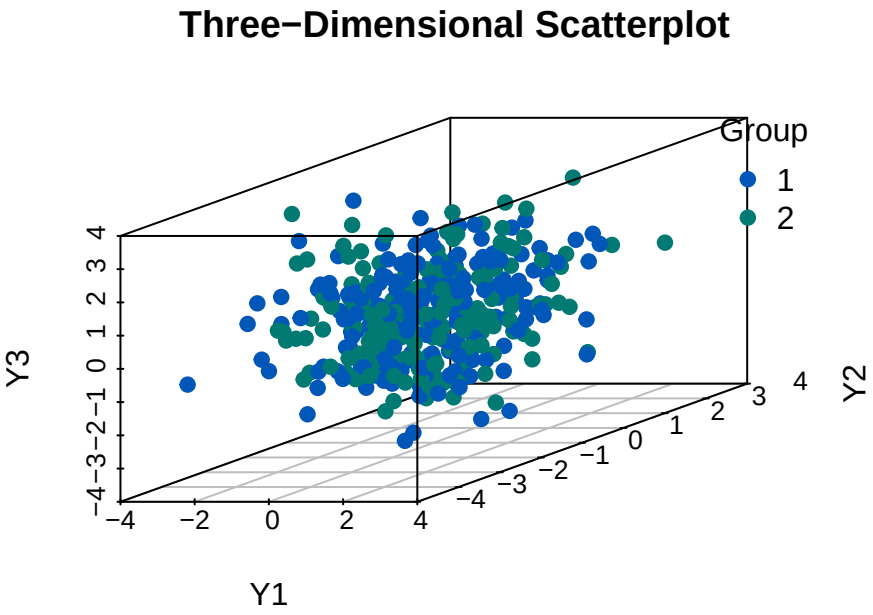


Figure 3: 3D Scatterplot of Independence from Cyclic Data

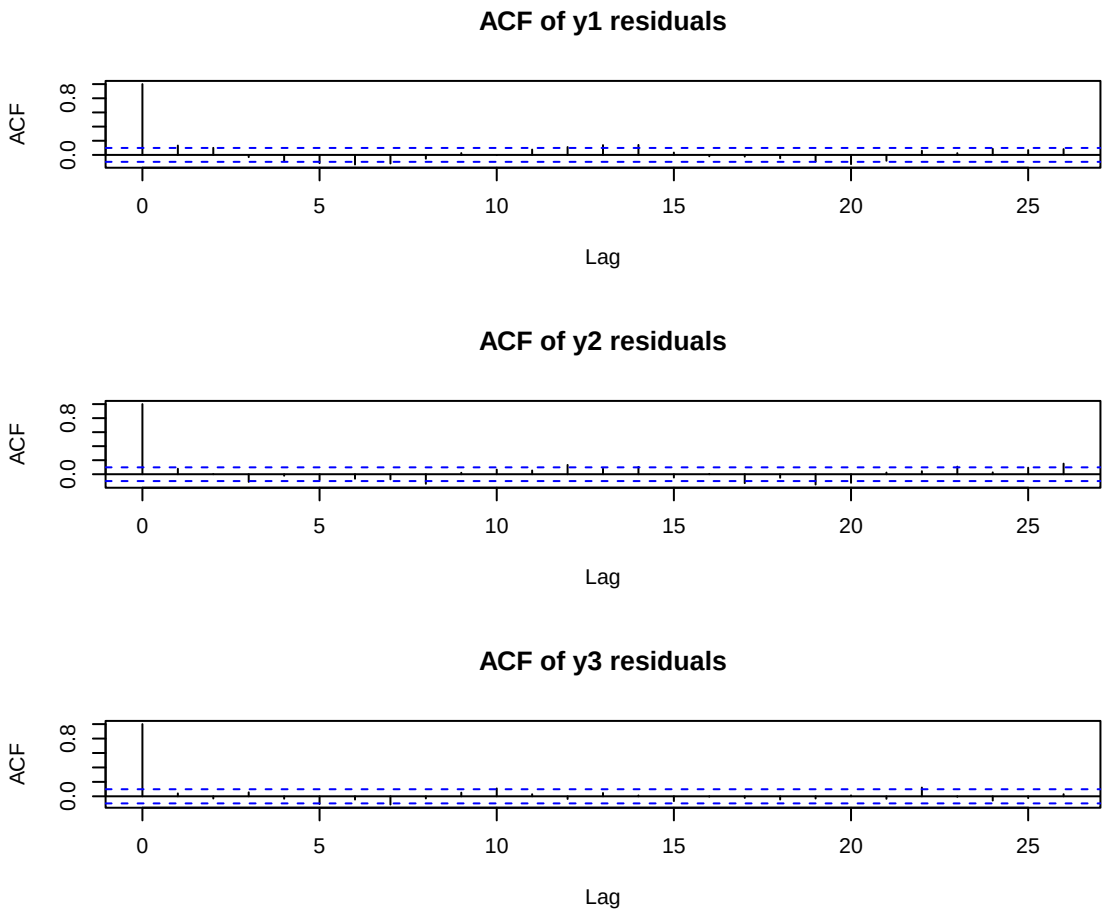


Figure 4: ACF Plot of Each Dimension's Residuals for Independence from Cyclic Data

3 Results

Table 1: True Difference in Means Trial Results (Expected: Reject Null)

Metric	**Proportion**
Matching results	23.8%
Matching fail to reject	23.8%
Matching reject	0.0%
MANOVA reject	76.2%
ANOVA majority reject	0.0%
Mismatched results	76.2%
True Y_1 value	100.0%
True Y_2 value	100.0%
True Y_3 value	100.0%

In the first experiment involving a true difference in means with an independence violation, MANOVA undoubtedly returned better results (Table~1). Although each univariate ANOVA successfully captured the true difference in means parameter for each dimension in all of the trials, the majority decision never rejected the null hypothesis. On the other hand, MANOVA acknowledged the true difference in means for 76% of the trials. In this case, MANOVA is more promising for detecting a small population difference within cyclic data.

Table 2: Independence from Cyclic Data Trial Results (Expected: Fail to Reject Null)

Metric	**Proportion**
Matching results	99.3%
Matching fail to reject	99.3%
MANOVA fail to reject	99.3%
ANOVA majority fail to reject	100.0%
Matching reject	0.0%
Mismatched results	0.7%
True Y_1 value	98.2%
True Y_2 value	98.6%
True Y_3 value	98.4%

For the second experiment with achieving independence from same distribution cyclic data, the results of both methods were comparable (Table~2). In this setting, ANOVA confidence intervals for the true difference in means contained the true parameter roughly 98% of the time, instead of 100% as it was previously. ANOVA failed to reject the null hypothesis in every trial while MANOVA failed to reject 99.3% of the time. Since the added normal noise had a sizable standard deviation and the population scatterplots are entirely overlapped, it is expected that both methods were able to easily detect no difference in means.

4 Discussion

When considering the aggregated outcomes of 1000 trials for each research setting, neither MANOVA nor ANOVA outperformed the other in entirety. When there is a true difference in means but an independence violation, MANOVA was clearly the superior technique. A shortcoming in our first experiment would be the difficulty in obtaining a majority decision; differences in mean vectors may be more apparent if there are additional populations in the simulation. In the second scenario, when populations had the same mean but enough noise added for data independence, both methods performed equally well. A weakness of this experiment is that the normally distributed noise may contribute a larger signal than the cyclic

data. Future areas of study could involve larger scale data with only minor mean differences or additional errors introduced in a portion of dependent vectors. It would also be valuable to explore differing cycle lengths between populations can affect the corresponding behavior of MANOVA.

References

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Landler, Lukas, Graeme D. Ruxton, and Eva P. Malkemper. 2022. "The Multivariate Analysis of Variance as a Powerful Approach for Circular Data." *Movement Ecology* 10 (1): 21. https://doi.org/10.1186/s40462-022-00323-8.

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